5 (**long**)

Figure 5 shows a pin jointed planar truss. The nodes of the truss are effectively restrained against out-of-plane movement. The members of the truss have cross-sectional area A and Young's modulus E. A vertical point load P is applied at joint H.

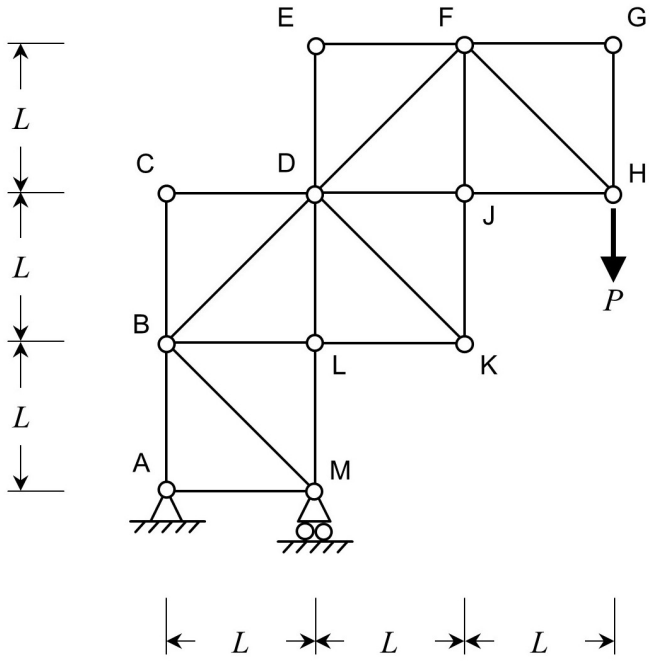


Fig. 5

(a) Determine the member forces in the truss in terms of P. [10]

(b) Determine the horizontal displacement of joint K in terms of P, L, A and E. [10]

(c) If L = 4 m and a steel circular hollow section of outside diameter 101.6 mm, wall thickness 4.0 mm, yield strength σ_y = 275 MPa and Young's modulus E = 200 GPa is selected for the design of all members, what is the maximum value of P that the truss can sustain elastically? [10]